

Anya Agarwal

anya04@seas.upenn.edu | 510-579-0092 | <https://anya0402.github.io/>

Education

University of Pennsylvania School of Engineering & Applied Science

December 2026

M.S.E in Computer Graphics and Game Technology

University of California, Berkeley College of Computing, Data Science, and Society

May 2025

B.A. in Computer Science, Certificate in Design Innovation

GPA: 3.86

Relevant Courses: Computer Graphics, GPU Programming, Computer Animation, 3D Computer Modeling, Data Structures, Algorithms, Signals and Systems, Machine Learning, Linear Algebra and Differential Equations, Operating Systems, Computer Architecture, User Interfaces, Probability, Data Science, Prototyping and Fabrication, Design Methodology

Skills

Python, Java, C, C++, Rust, Pandas, SQL, Numpy, HTML, Javascript, Typescript, Tailwind CSS, React.js, Adobe Photoshop & After Effects, Autodesk Maya, Figma

Experience

Berkeley Institute of Design

September 2024 - May 2025

Undergraduate Researcher

- Assisted in conducting and writing a research paper about the creation of Noise Pilot — a tool that implements diffusion-based image generation in a node-based image processing interface — and its impact and usability for artists.
- Connected Noise Pilot with a drawing interface API to allow users to draw masks and shapes for their image generation
- Implemented an inverse kinematics algorithm from a research paper to explore movement in VR environments.

LearnQ.ai

June 2023 - August 2023

Fullstack Software Engineer Intern

- Implemented a complete UI for new AI chatbot feature and menu page using React Typescript and Tailwind CSS.
- Generated several AI chatbot modes using LangChain and OpenAI, and trained them on varying knowledge bases.
- Designed a framework for the chatbot to handle connections between frontend and backend modules.

Berkeley Data Science Department

June 2023 - August 2024

Undergraduate Student Instructor for Intro to Data Science and Advanced Data Science Topics

- Taught weekly discussion section (20+ students) to review lecture content, walk through a discussion-based problem set.
- Held tutor sections and office hours to guide students through data science topics and answer conceptual questions.
- Managed course infrastructure — used GitHub to upload assignments, and debugged students' JupyterHub bugs.

Projects

Cloth Simulator

April 2024

- Utilized mass and spring system to create a cloth simulation and its corresponding collisions with other objects.
- Created Blinn-Phong, bump, displacement, texture, and other shaders for the cloth using the OpenGL API.

Pathtracer

March 2024

- Used the bounding volume hierarchy algorithm to accelerate ray intersections with triangles and spheres.
- Rendered images using global illumination (direct/indirect lighting), and optimized using Monte Carlo probability.

Honors/Awards

Grace Hopper Celebration Student Scholarship

June 2023

- Scholarship for women and non-binary undergraduates to attend the Grace Hopper Celebration and participate in various networking and skill building workshops.